PRIYANRAJ J

AI & Embedded Systems Engineer | Data Science Student

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Portfolio: priyanrajj-hub.github.io/PORTFOLIO

SUMMARY

First-year engineering student pursuing three concurrent degrees, with hands-on project work across AI-driven web applications, low-cost sensing hardware, and systems programming in C/C++. Regular hackathon participant and competitive programmer, focused on building working prototypes and turning them into research.

EDUCATION

Amrita Vishwa Vidyapeetham, Chennai - B.Tech, Computer and Communication Engg. 2025 - 2029

Indian Institute of Technology Madras - BS, Data Science Ongoing

Indian Institute of Technology Guwahati - BS, Data Science and Artificial Intelligence Ongoing

PROJECTS

Canopy - Global Vegetation Health Monitor | React, Firebase, Gemini API, Sentinel imagery, Open-Meteo, Vercel

- Web app where users draw a field polygon on a map and receive vegetation analysis from satellite imagery and live weather telemetry.
- Integrated Gemini-generated insights for plant condition, disease and pest risk; selected for project review as a software solution, with capacitive hardware integration planned.

Low-Cost Capacitive Plant Water-Stress Sensing | ESP32, Arduino, FDC1004, DS18B20, Python

- Smart India Hackathon capstone proposing a capacitive dielectric leaf probe as a low-cost alternative to THz spectroscopy for early stress detection.
- Built a multi-sensor ESP32 prototype (DHT11, light, gas, rain, ultrasonic) with LCD readout; designed a 3-4 week tomato/chili trial protocol and published a live dashboard.

RoadWatch and DriveLegal | Python

- Developed RoadWatch (road quality monitoring) and DriveLegal (AI-powered traffic-law assistant for India) for the National Road Safety Hackathon 2026 at IIT Madras.

EduSense AI | Claude API

- Built an AI study assistant for a Build-A-Thon on Unstop.

Pharmaceutical Evergreening Detection System | C/C++

- Patent-registry program using pointers, structs and file I/O, with Levenshtein-based fuzzy drug-name matching and CSV/log export.

Friend Recommendation using Graphs (Research Exploration) | Graph algorithms, React, Vercel

- Studied LinkedIn-style recommendation via BFS N-hop, Personalized PageRank, Jaccard, Adamic-Adar and embeddings; built an interactive explainer site.

HACKATHONS AND COMPETITIONS

- RIT Quant-A-Thon 2026 (QT-1.6): MATLAB simulation of quantum entanglement and correlation statistics.
- National Road Safety Hackathon 2026 (IIT Madras): RoadWatch, DriveLegal. | Sloka Hackathon 2026: Chakravyuh, women's safety platform.
- Smart India Hackathon: capacitive plant stress sensing project. | Active on LeetCode, Codeforces, AtCoder, HackerRank.

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript, TypeScript, MATLAB, HTML/CSS
AI and Data: Gemini API, Claude API, graph algorithms
Web and Tools: React, Vite, Tailwind CSS, Firebase, Vercel, Git/GitHub, ReportLab
Hardware: ESP32, Arduino, capacitive and environmental sensors

MEMBERSHIPS AND ACTIVITIES

- IEEE Active Member; CAMHI Member.
- Designed ID cards and the TechQuest event flyer for the IEEE CASS and SPS student chapters at Amrita.